Ablations and control baselines. To test necessity, we perform SAE-based ablations by encoding the decision-state activation $x^{(\ell)}$, zeroing the selected refusal latents in $z^{(\ell)}$, and decoding back into residual space to obtain an ablated activation $\tilde{x}^{(\ell)} = g_\phi(\tilde{z}^{(\ell)})$ that is used in place of $x^{(\ell)}$ during generation. As controls, we construct steering directions from random latent subsets of the same size K, and from random unit vectors in residual space.

3 Data and Models

To capture a broad range of refusal and non-compliance behaviors, we draw on four datasets with complementary characteristics, from which we construct 11 evaluation splits.

WildGuardMix (WGM) provides ground-truth safety annotations indicating whether prompts should be complied with or declined under safety policies; from this dataset we construct the **SafetyCore–WGM** split containing both safety-sensitive and benign prompts.

SorryBench (SB) organizes unsafe content into four policy domains, from which we construct four splits corresponding to hate speech generation (**HateSpeech–SB**), potentially inappropriate topics (**Inappropriate–SB**), assistance with crimes or torts (**CrimeAssistance–SB**), and unqualified professional advice (**Advice–SB**).

CoCoNot (CCN) covers contextual non-compliance beyond conventional safety cases; we construct five splits capturing incomplete requests (**Incomplete–CCN**), unsupported requests (**Unsupported–CCN**), indeterminate requests (**Indeterminate–CCN**), humanizing requests (**Humanizing–CCN**), and safety-related requests involving harmful content (**Safety–CCN**).

XSTest (XST) targets over-refusal by contrasting adversarially framed but benign prompts with standard benign requests; we construct the **OverRefusal–XST** split to study inappropriate refusal on safe inputs. Additional dataset details are provided in Appendix B.

Each evaluation split consists of a balanced pair of prompt sets: at least 32 prompts labeled to elicit non-compliant behavior under dataset annotations, paired with an equal number of benign prompts for which compliance is appropriate. Following Arditi et al. (2022), we adopt 32 prompts per class as a standard unit for estimating stable activation centroids. When sufficient data is available, we additionally construct multiple independent 32/32 splits without replacement, enabling validation of within-category geometric stability.

Models: We run all experiments on two instruction-tuned models: `gemma-2-9b-it`[2] and `Llama-3.1-8B-Instruct`[3]. Activation-space directions are computed from residual-stream activations at a fixed mid-layer hook (`resid_pre`; e.g., layer 20 for Gemma and layers 15–16 for Llama). For sparse-feature analyses, we use publicly available residual-stream SAEs (GemmaScope layers 9/20/31 and the `saes-llama-3.1-8b-instruct` release[4]), focusing on later layers where refusal-related features are most pronounced. Activations are extracted at the chat-template token immediately preceding the assistant's response (index -2), which we treat as the model's decision state.

4 Findings

4.1 Refusal Directions in Activation Space

In this section, we analyze refusal behavior directly in activation space. For each of the 11 refusal splits, we compute a corresponding refusal direction and quantify similarity across splits using pairwise cosine distance. As shown in Table 1, many directions are substantially dissimilar (typical cosine similarity 0.4–0.6, with several near-orthogonal), indicating that different refusal categories correspond to distinct activation-space directions.

Controlled test: To evaluate whether these directions causally mediate refusal, we use a controlled test set of 200 prompts balanced across prompt type (harmful vs. benign) and the unsteered model's response (refusal vs. compliance). This yields four equally sized subsets: HR, HC, BR, and BC. By construction, the unsteered model achieves 50% accuracy, since only refusals to harmful prompts (HR) and compliance with benign prompts (BC) are correct outcomes; HC and BR correspond to jailbreaks and over-refusals, respectively.

Steering: We steer the base model along each of the 11 refusal directions and evaluate performance on the controlled test set. Steering is applied at various strengths, ranging from weak to near-saturated

[2] `hf.co/google/gemma-2-9b-it`
[3] `hf.co/meta-llama/Llama-3.1-8B-Instruct`
[4] `hf.co/andyrdt/saes-llama-3.1-8b-instruct`

	SafetyCore WGM	OverRefusal XST	Humanizing CCN	Incomplete CCN	Indeterminate CCN	Safety CCN	Unsupported CCN	HateSpeech SB	CrimeAssist. SB	Inappropriate SB	Advice SB
SafetyCore-WGM	1.000	0.632	0.234	0.127	0.310	0.663	0.297	0.695	0.734	0.649	0.511
OverRefusal-XST	0.632	1.000	0.239	-0.062	0.203	0.548	0.099	0.609	0.715	0.390	0.426
Humanizing-CCN	0.234	0.239	1.000	0.460	0.678	0.586	0.620	0.417	0.411	0.328	0.474
Incomplete-CCN	0.127	-0.062	0.460	1.000	0.688	0.550	0.627	0.415	0.402	0.379	0.468
Indeterminite-CCN	0.310	0.203	0.678	0.688	1.000	0.719	0.795	0.560	0.573	0.502	0.666
Safety-CCN	0.663	0.548	0.586	0.550	0.719	1.000	0.692	0.876	0.885	0.822	0.811
Unsupported-CCN	0.297	0.099	0.620	0.627	0.795	0.692	1.000	0.499	0.480	0.520	0.619
HateSpeech-SB	0.695	0.609	0.417	0.415	0.560	0.876	0.499	1.000	0.917	0.862	0.746
CrimeAssistance-SB	0.734	0.715	0.411	0.402	0.573	0.885	0.480	0.917	1.000	0.809	0.795
Inappropriate-SB	0.649	0.390	0.328	0.379	0.502	0.822	0.520	0.862	0.809	1.000	0.732
Advice-SB	0.511	0.427	0.474	0.468	0.666	0.811	0.619	0.746	0.795	0.732	1.000

Table 1: Pairwise cosine similarity matrix between refusal directions learned from the 11 evaluation splits. The direction of SafetyCore-WGM is the closest to the safety refusal direction of (Arditi et al., 2022).
WGM: WildGuard-Mix, CCN: CoCoNot, SB: SorryBench, XST: XSTest

refusal regimes as in Table 12. Steered responses are labeled as refusals or compliant answers using WildGuard, with manual validation. We report overall accuracy, refusal rate, and over-refusal rate.

Table 2 reports summary results for steering and direction ablation of a safety-derived direction for Gemma and Llama. Across splits and models, steering along different refusal directions produces broadly similar behavior: increasing steering strength raises refusal rates on harmful prompts while simultaneously increasing over-refusal on benign prompts, with both approaching saturation at high strength. A notable exception is CoCoNot non-safety categories which respond more gradually to steering. Direction ablation for Gemma shows the complementary effect, suppressing refusal in most cases, though CoCoNot-derived directions leave residual refusal rates of approximately 0.3–0.5, suggesting partial misalignment with safety refusal mechanisms. Also, Llama does not respond well to the safety direction ablation, suggesting a richer internal refusal structure with redundant paths that survived the ablation.

Despite these geometric differences, steered models consistently refuse to answer prompted questions, differing primarily in *how* refusal is expressed. As illustrated in Appendix A, humanizing directions emphasize the model's non-human nature, incomplete-request directions produce terse expressions of confusion, and indeterminate or unsupported directions stress impossibility or lack of agency. SorryBench directions foreground moral judgments or professional disclaimers, while WildGuard and XSTest directions emphasize safety, illegality, or harm.

Overall, these results indicate that refusal performance is largely invariant across directions, while refusal *style* varies substantially, consistent with refusal acting as a simple behavioral switch with stylistic degrees of freedom.

4.2 Refusal Directions in SAE Space

To probe refusal behavior at the level of internal features, we analyze refusal in the latent space of sparse autoencoders (SAEs). For the 11 data splits, we identify refusal-associated SAE latents following Section 2.2 and examine: (i) their behavior under causal intervention, (ii) their alignment with the activation-space refusal directions from Section 4.1, (iii) the extent to which refusal latents are reused across datasets, splits, and layers, and (iv) the semantic themes these latents encode.

4.2.1 Latent-Space Steering

We test whether the refusal-associated SAE latents are *causally involved* in refusal behavior, by asking whether steering in latent space recovers the same one-dimensional refusal knob observed in activation space. For each refusal split, we construct a single SAE-based refusal direction by averaging the decoder directions of the top-ranked refusal latents (Section 2.2), then apply residual-stream steering for the same controlled test set from Section 4.1 during generation at layer ℓ, $r' = r + \beta\, d_{\mathrm{SAE}}^{(s,\ell)}$. We choose the steering strength β using a coarse grid search on the validation pool (Gemma: $\beta \in \{10, 30, 60\}$; Llama: $\beta \in \{0.5, 0.8, 1.0, 1.2\}$). WildGuard is used again to label steered refusals R' or compliance C'.

Table 3 summarizes the performance of SAE-based steering across splits for Gemma and Llama (the underlying safety-tuned base model at $\beta = 0$ already exhibits substantial refusal and over-refusal). Across all splits, increasing β monotonically raises refusal rates on harmful prompts and, in parallel, over-refusal rates on benign prompts, tracing out essentially the same accuracy–over-refusal trade-off observed when steering directly in activa-

Split	Gemma (**steered,** $\alpha = 100$)			Llama (**steered,** $\alpha = 2.5$)			Gemma (**ablated**)			Llama (**ablated**)		
	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.515	1.00	0.97	0.515	0.880	0.850	0.723	0.45	0.00	0.819	0.64	0.00
Incomplete-CCN	0.550	0.98	0.88	0.475	0.650	0.700	0.670	0.34	0.00	0.650	0.46	0.16
Indeterminate-CCN	0.545	0.99	0.90	0.520	0.970	0.930	0.713	0.45	0.02	0.610	0.40	0.18
Safety-CCN	0.500	1.00	1.00	0.510	1.000	0.980	0.574	0.15	0.00	0.595	0.48	0.29
Unsupported-CCN	0.515	0.99	0.96	0.510	0.920	0.900	0.723	0.45	0.00	0.585	0.48	0.31
CrimeAssistance-SB	0.500	1.00	1.00	0.500	1.000	1.000	0.500	0.00	0.00	0.510	0.42	0.40
HateSpeech-SB	0.500	1.00	1.00	0.500	1.000	1.000	0.500	0.00	0.00	0.525	0.43	0.38
Inappropriate-SB	0.500	1.00	1.00	0.505	0.990	0.980	0.500	0.00	0.00	0.575	0.48	0.33
Advice-SB	0.500	1.00	1.00	0.500	1.000	1.000	0.532	0.06	0.00	0.545	0.35	0.26
SafetyCore-WGM	0.500	1.00	1.00	0.495	0.990	1.000	0.553	0.11	0.00	0.515	0.38	0.35
OverRefusal-XST	0.510	1.00	0.98	0.500	1.000	1.000	0.521	0.04	0.00	0.570	0.63	0.49

Table 2: Performance of Gemma and Llama models **steered** to refuse, or **ablated** to comply, across 11 refusal directions on the controlled test set, reported in terms of accuracy: $Acc = (HR' + BC')/200$, refusal rate: $RR = HR'/100$, and over-refusal rate: $ORR = BR'/100$, where H and B refer to harmful and benign prompts, respectively, and R' and C' refer to the WildGuard judgment of whether the steered response is a refusal or a compliance, respectively. The unsteered base models attain 50% on all three metrics.

tion space. This shows that steering along a small SAE-derived direction is sufficient to recover the same one-dimensional refusal knob.

Control experiments using a random subset of SAE latents confirm that this behavior is specific to the identified refusal latents, produceing much weaker, non-monotonic changes in refusal.

4.2.2 Alignment with Activation-Space Refusal Directions

To quantify if activation-space refusal directions can be reconstructed from the small set of refusal-associated SAE latents, for each prompt we obtain the SAE latent activations at the decision state $x^{(\ell)}$, filter for the top-ranked refusal latents (Section 2.2), reconstruct their contribution in residual space by summing the corresponding decoder directions weighted by the latent activations, and compute the cosine similarity between this reconstructed direction and the corresponding activation-space direction. We report the mean and standard deviation of the cosine similarity over prompts for each split in Table 4. Cosine similarity between SAE directions and activation-space directions is uniformly high across datasets and refusal categories, typically in the range 0.85–0.97, indicating good reconstruction. Despite the fact that activation-space refusal directions are geometrically distinct across datasets (Section 4.1), they appear to operate on substantially shared underlying SAE features; this might help explain why steering along different activation-space refusal directions produces similar refusal efficacy.

4.2.3 Latent Overlaps Across Dataset Splits

To assess how much of the SAE refusal structure is shared across datasets and splits, we rank latents within each split by the difference in their firing rates on high-risk (HR) versus base (BC) subsets and retain the top 1000 latents per split.

For each layer, we count (i) how many distinct latents appear in at least one of these lists, and (ii) the latents that appear in for *all* the 11 splits. These overlap statistics are summarized in Table 5, while detailed latent–split incidence matrices and top-shared-latent tables are given in Appendix F.

As the results show, the signal is *sparse but consistent*: in all three layers, a relatively small subset of latents carries most of the HR vs. BC discrimination signal with only about 8–10% and 2.5–4% of the 16,384 latents ever appear in any top–1000 list or in all splits, respectively.

Secondly, there are *robust cores* of refusal latents within each layer with strict intersections of 591, 517, and 421 latents for layers 9, 20, and 31, respectively.

Third, there is a *depth trend*: the fraction of latents that ever appear in a top–1000 list grows slightly with depth (from ∼8.55% at layer 9 to ∼9.64% at layer 31), while the size of the strict core shrinks (from ∼3.61% to ∼2.57%). This suggests that deeper layers distribute the HR vs. BC signal over a somewhat larger basin, with a smaller, more specialized subset that is truly uni-